

Q1. Create a Swing form to enter the guest details who visits the LNMIIT guest house. The following details may be expected from the guest at the time of check-in: Name, address, id-proof code, mobile number, Gender, purpose of visit along with check-in date and checkout date and room number. The check-in date should be automatically filled as the system date and should not be editable. Apply all validations such as name should contain only alphabets and mobile number should contain 10 digits. There are various types of rooms in the guest house and is allotted on the basis of availability. The Swing form should also contain a button to view the details of any Guest on the basis of Name or Mobile number. If search is on the basis of “name” then the Swing form should have previous and next buttons to navigate backward and forward record.

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import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.*;

// ----- Guest Class -----

class Guest {

    private String name, address, idProof, mobile, gender, purpose, checkInDate,
    checkOutDate, roomNo;

    public Guest(String name, String address, String idProof, String mobile, String
    gender,

        String purpose, String checkInDate, String checkOutDate, String roomNo)
    {

        this.name = name;
        this.address = address;
        this.idProof = idProof;
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        this.mobile = mobile;

        this.gender = gender;

        this.purpose = purpose;

        this.checkInDate = checkInDate;

        this.checkOutDate = checkOutDate;

        this.roomNo = roomNo;
    }

    // Getters

    public String getName() { return name; }

    public String getAddress() { return address; }

    public String getIdProof() { return idProof; }

    public String getMobile() { return mobile; }

    public String getGender() { return gender; }

    public String getPurpose() { return purpose; }

    public String getCheckInDate() { return checkInDate; }

    public String getCheckOutDate() { return checkOutDate; }

    public String getRoomNo() { return roomNo; }

    // To display guest info

    @Override

    public String toString() {

        return "Name: " + name + "\nAddress: " + address + "\nID Proof: " + idProof +
"\nMobile: " + mobile +

            "\nGender: " + gender + "\nPurpose: " + purpose + "\nCheck-In: " +
checkInDate +

            "\nCheck-Out: " + checkOutDate + "\nRoom No: " + roomNo;

    }

}

// ----- GuestHouse Class -----

class GuestHouse {

    private ArrayList<Guest> guestList = new ArrayList<>();

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public void addGuest(Guest g) {
    guestList.add(g);
}

public ArrayList<Guest> searchByName(String name) {
    ArrayList<Guest> result = new ArrayList<>();
    for (Guest g : guestList) {
        if (g.getName().equalsIgnoreCase(name)) {
            result.add(g);
        }
    }
    return result;
}

public Guest searchByMobile(String mobile) {
    for (Guest g : guestList) {
        if (g.getMobile().equals(mobile)) return g;
    }
    return null;
}

public ArrayList<Guest> getGuestList() {
    return guestList;
}
}

// ----- GuestForm Class -----

public class GuestForm extends JFrame implements ActionListener {

    private JTextField nameField, addressField, idField, mobileField, purposeField,
    checkInField, checkOutField, roomField, searchField;

    private JComboBox<String> genderBox;

    private JButton addBtn, searchBtn, nextBtn, prevBtn;

    private GuestHouse guestHouse;

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private ArrayList<Guest> searchResults;

private int currentIndex = 0;

private JRadioButton searchByName, searchByMobile;

public GuestForm() {

    setTitle("LNMIIT Guest House Management");

    setSize(600, 500);

    setLayout(new GridLayout(12, 2, 5, 5));

    setDefaultCloseOperation(EXIT_ON_CLOSE);

    setLocationRelativeTo(null);

    guestHouse = new GuestHouse();

    // Form fields

    nameField = new JTextField();

    addressField = new JTextField();

    idField = new JTextField();

    mobileField = new JTextField();

    genderBox = new JComboBox<>(new String[]{"Male", "Female", "Other"});

    purposeField = new JTextField();

    SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");

    checkInField = new JTextField(sdf.format(new Date()));

    checkInField.setEditable(false);

    checkOutField = new JTextField();

    roomField = new JTextField();

    // Buttons

    addBtn = new JButton("Add Guest");

    searchBtn = new JButton("Search Guest");

    nextBtn = new JButton("Next");

    prevBtn = new JButton("Previous");

    // Search options

    searchByName = new JRadioButton("Search by Name");

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searchByMobile = new JRadioButton("Search by Mobile");
ButtonGroup bg = new ButtonGroup();
bg.add(searchByName);
bg.add(searchByMobile);
searchField = new JTextField();
// Add components
add(new JLabel("Name:"));
add(nameField);
add(new JLabel("Address:"));
add(addressField);
add(new JLabel("ID Proof Code:"));
add(idField);
add(new JLabel("Mobile Number:"));
add(mobileField);
add(new JLabel("Gender:"));
add(genderBox);
add(new JLabel("Purpose of Visit:"));
add(purposeField);
add(new JLabel("Check-In Date:"));
add(checkInField);
add(new JLabel("Check-Out Date:"));
add(checkOutField);
add(new JLabel("Room Number:"));
add(roomField);
add(addBtn);
add(new JLabel("---- Search Section ----"));
add(searchByName);
add(searchByMobile);
add(new JLabel("Search Value:"));
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        add(searchField);
        add(searchBtn);
        add(prevBtn);
        add(nextBtn);
        // Action Listeners
        addBtn.addActionListener(this);
        searchBtn.addActionListener(this);
        nextBtn.addActionListener(this);
        prevBtn.addActionListener(this);
        prevBtn.setEnabled(false);
        nextBtn.setEnabled(false);

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == addBtn) {
            addGuest();
        } else if (e.getSource() == searchBtn) {
            searchGuest();
        } else if (e.getSource() == nextBtn) {
            showNextGuest();
        } else if (e.getSource() == prevBtn) {
            showPreviousGuest();
        }
    }

    private void addGuest() {
        String name = nameField.getText().trim();
        String address = addressField.getText().trim();
    }

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String id = idField.getText().trim();
String mobile = mobileField.getText().trim();
String gender = genderBox.getSelectedItem().toString();
String purpose = purposeField.getText().trim();
String checkIn = checkInField.getText().trim();
String checkOut = checkOutField.getText().trim();
String room = roomField.getText().trim();

// Validation
if (!name.matches("[a-zA-Z ]+")) {
    JOptionPane.showMessageDialog(this, "Name must contain only
alphabets!");
    return;
}
if (!mobile.matches("\\d{10}")) {
    JOptionPane.showMessageDialog(this, "Mobile number must be 10 digits!");
    return;
}

Guest guest = new Guest(name, address, id, mobile, gender, purpose, checkIn,
checkOut, room);
guestHouse.addGuest(guest);
JOptionPane.showMessageDialog(this, "Guest added successfully!");
clearFields();
}

private void searchGuest() {
    String searchValue = searchField.getText().trim();
    if (searchByName.isSelected()) {
        searchResults = guestHouse.searchByName(searchValue);
        if (searchResults.isEmpty()) {
            JOptionPane.showMessageDialog(this, "No guest found with that name.");
            return;
        }
    }
}

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    }

    currentIndex = 0;

    displayGuest(searchResults.get(currentIndex));

    updateNavButtons();
} else if (searchByMobile.isSelected()) {

    Guest g = guestHouse.searchByMobile(searchValue);

    if (g == null) {

        JOptionPane.showMessageDialog(this, "No guest found with that mobile
number.");

    } else {

        displayGuest(g);

    }

    nextBtn.setEnabled(false);

    prevBtn.setEnabled(false);

} else {

    JOptionPane.showMessageDialog(this, "Please select a search option.");

}

}

private void displayGuest(Guest g) {

    JOptionPane.showMessageDialog(this, g.toString(), "Guest Details",
JOptionPane.INFORMATION_MESSAGE);

}

private void showNextGuest() {

    if (searchResults != null && currentIndex < searchResults.size() - 1) {

        currentIndex++;

        displayGuest(searchResults.get(currentIndex));

    }

    updateNavButtons();

}

private void showPreviousGuest() {

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        if (searchResults != null && currentIndex > 0) {
            currentIndex--;
            displayGuest(searchResults.get(currentIndex));
        }
        updateNavButtons();
    }

    private void updateNavButtons() {
        if (searchResults == null || searchResults.size() <= 1) {
            nextBtn.setEnabled(false);
            prevBtn.setEnabled(false);
            return;
        }
        prevBtn.setEnabled(currentIndex > 0);
        nextBtn.setEnabled(currentIndex < searchResults.size() - 1);
    }

    private void clearFields() {
        nameField.setText("");
        addressField.setText("");
        idField.setText("");
        mobileField.setText("");
        purposeField.setText("");
        checkOutField.setText("");
        roomField.setText("");
    }

    public static void main(String[] args) {
        new GuestForm();
    }
}

```